Maya v2: how much power to buy the day before

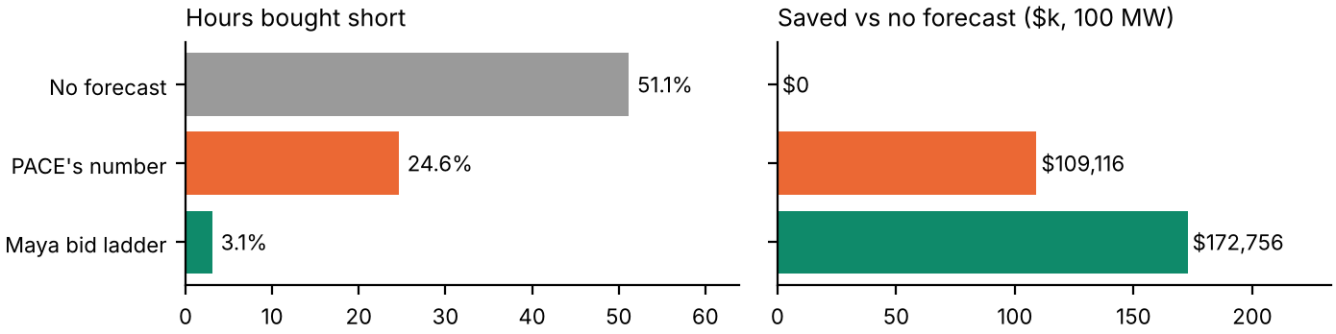
Maui Hiva Innovations | Purdue MS Data Science capstone, Project T | updated October 20, 2026 | live dashboard: maya-v2-buy-ahead.streamlit.app

What it does

Every morning before the 10 am Mountain day-ahead deadline, Maya forecasts tomorrow's net load for the PacifiCorp East area (demand minus Milford, Utah renewable output) as a range, not a single number, and turns it into a buy plan. Buying short means paying the real-time price for the gap, so the plan is built to be short rarely and on purpose. It also flags hours at risk of a price spike, suggests when to run flexible load, and gives a 7-day outlook of daily net load for weekly planning.

Results

	Maya	Comparison
Best-guess error, 2026 (MWh per hour)	**229**	PACE's own forecast: 343 (33% higher)
Live, 48 unseen days: hours bought short	**3.4%**	PACE's number: 30%
Live best-guess error (MWh per hour)	**302**	PACE: 480
Saved vs buying with no forecast*	**$172,756**	90% range $68,999 to $288,933
Saved vs buying PACE's number*	**$63,640**	90% range $19,424 to $108,058
Load shifting, 5 MW moved 4 h a day*	**$88,862**	92% of the perfect-hindsight value; saved money on 100% of days



*Real CAISO day-ahead and real-time prices for the PACE area, May 01 to Oct 20, 2026 (171 days), at a placeholder size of 100 MW until Maui Hiva confirms its load. 90% ranges from a bootstrap over whole days.

How it works

1. A calibrated range. Two models (gradient boosting and an LSTM) forecast levels from P50 (best guess) to P98, using ECMWF weather forecasts and only data available at the deadline. Each level is re-checked daily, so P95 really is short about 5% of the time.

2. Risk and the bid ladder. A daily Green / Yellow / Red tier sets the always-bought amount (P95, or P98 on risky days). An hourly spike-risk model sets a price limit for an extra block up to P98, bought only when day-ahead is cheaper than expected real-time.

3. Load shifting. Expected hourly prices (last 7 days of day-ahead results) show which hours to move flexible load out of and into.

4. Runs itself. A daily job pulls EIA demand, ECMWF weather and CAISO prices, forecasts, and scores the track record. A monthly check retrains the models and switches only if the new one is better on unseen days.

Limits

Day-ahead prices for PACE exist only since May 2026 (CAISO EDAM), so dollar results rest on a short window. Milford renewable output is modeled, not metered. Maui Hiva's own load is assumed to follow PACE's pattern. On ordinary days the buy strategies differ by cents per MWh; Maya's main value is being short far less often, which is what matters on spike days.

Dashboard: maya-v2-buy-ahead.streamlit.app | Code and data: github.com/RendellRich/maya-forecasting-v2